Linking Manufacturing Defects to Operating Conditions and Causal Evidence

Capstone Engineering Team (Members 1–4) • Project Planning & Management Proposal

Abstract—*In modern discrete manufacturing lines, automated optical inspection identifies surface flaws but fails to deduce underlying physical causation. An observed surge in defective parts may stem from raw material batch anomalies, machine tool degradation, drift in operating setpoints, or purely external optical sensor artifacts. We present a causal manufacturing intelligence platform that pairs KolektorSDD2 surface defect segmentation with 100 Hz high-frequency machine telemetry via Managed Service for Apache Flink temporal interval joins. To resolve mathematical non-identifiability between optical measurement shifts and genuine process wear, we enforce Requirement R-CAM (dedicated optical health telemetry) and apply Rogan–Gladen prevalence corrections. AWS Glue lakehouse analytics and Amazon Bedrock Claude 3.7 Sonnet agent reasoning with strict event-time leakage guards deliver automated, evidence-verified causal attributions across eight benchmark scenario families.*

Index Terms—*Causal Inference, Industrial Computer Vision, Stream Processing, KolektorSDD2, Rogan–Gladen Estimator, Bedrock Claude 3.7 Sonnet, MLOps.*

I. PROBLEM FORMULATION, DECOUPLING & CAUSAL IDENTIFIABILITY

A. The Core Industrial Challenge & False Explanations

Automated surface inspection typically operates as an isolated classification or segmentation step, labeling discrete parts as 'good' or 'defective' without context. On a high-speed production line, however, a sudden increase in detected defect rates can stem from multiple, highly distinct mechanisms: raw material impurities, tool wear, setpoint drift, or external optical camera vibration. Treating defect detection as an unlinked classification task produces false explanations and misdirected maintenance. The unit of analysis must be the part and its precise spatiotemporal production context.

Furthermore, linking an image to sensor readings from another part or averaging sensor measurements over coarse hourly shifts obscures causal signals and introduces cross-part contamination. Discrete parts must be individually tracked through the production line using a deterministic identifier (part_id), with sensor windows strictly aligned to the exact transit duration through each machine station.

B. Mathematical Decoupling & The Rogan–Gladen Estimator

The central causal identifiability challenge is that an observed defect rate can be shifted by two fundamentally different phenomena: genuine changes in physical manufacturing quality, or shifts in the optical measurement system itself. Let p denote the true, latent defect prevalence in produced parts. The measured defect rate is governed by the model's true positive rate (TPR) and false positive rate (FPR):

$$\textit{Measured Defect Rate} = TPR \cdot p + FPR \cdot (1 - p)$$

While raw material batch changes and tool wear alter p, external camera vibration or lens focal drift shifts the segmentation model's TPR and FPR. From defect inspection counts alone, these causes are completely unidentifiable. To break this mathematical symmetry, we enforce Requirement R-CAM: a dedicated optical-health telemetry channel that tracks camera pose residuals, vibration metrics, and periodic in-situ golden-sample inspections. When golden samples provide empirical calibration of TPR and FPR, the true physical defect prevalence p is uniquely recovered via the Rogan–Gladen estimator:

$$p = (\textit{Measured Defect Rate} - FPR) / (TPR - FPR)$$

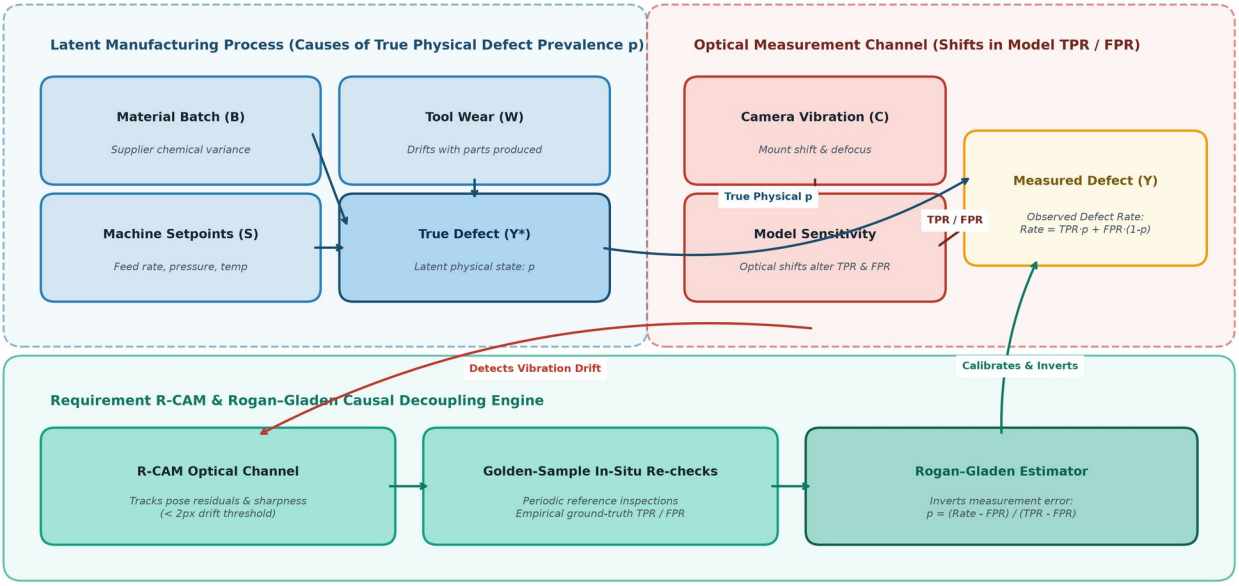


Fig. 1. *Structural Causal Model (SCM) DAG illustrating the decoupling between latent physical process causes (B, S, W) and optical measurement shifts (C) via Requirement R-CAM and Rogan–Gladen calibration.*

C. Scenario Family Matrix (S1–S8)

To rigorously evaluate causal reasoning and ensure the agent does not fabricate singular causes under confounded conditions, the platform is benchmarked against eight distinct causal scenario families:

TABLE I: CAUSAL SCENARIO FAMILY BENCHMARK MATRIX (S1–S8)

Scenario	Physical Mechanism	Telemetry & Image Manifestation	Expected Causal Attribution
S1: Batch Shift	Raw material batch variation	True prevalence p increases; sensor telemetry nominal; R-CAM nominal.	Material batch ranked #1 with tight confidence bounds.
S2: Camera Shift	Camera mount vibration/drift	p unchanged; model TPR/FPR shift; R-CAM detects pose residual > 2px.	Camera vibration ranked #1; Rogan-Gladen rate applied.
S3: Setting Drift	Machine feed rate/temp drift	Step/ramp in sensor readings; dose-response relationship with flaws.	Operating setpoint ranked #1; change-point isolated.
S4: Tool Wear	Progressive cutter wear	Flaw rate rises monotonically with parts produced; resets on maintenance.	Tool wear curve ranked #1; maintenance interval flagged.
S5: Confounded	Batch change & setpoint drift	Multiple collinear shifts occur simultaneously within the same shift.	Agent explicitly abstains from singular attribution; logs test.
S6: Spurious	Fluctuation in idle sensor	Ambient noise in unrelated telemetry channel; no physical correlation.	Distractor dismissed; no false causal link established.
S7: Data Fault	Clock skew or stream drop	Out-of-order timestamps, duplicate part_id, or schema anomaly.	Data-quality alert raised; causal inference halted.
S8: Null Baseline	Routine steady-state line	Routine stochastic Poisson noise; all parameters within control limits.	No causal anomaly reported; false discovery rate = 0%.

II. END-TO-END SYSTEM & CAUSAL DATAFLOW ARCHITECTURE

The end-to-end platform couples high-throughput real-time stream processing with an analytical lakehouse and an agentic causal reasoning layer. Figure 2 illustrates the complete four-stage dataflow:

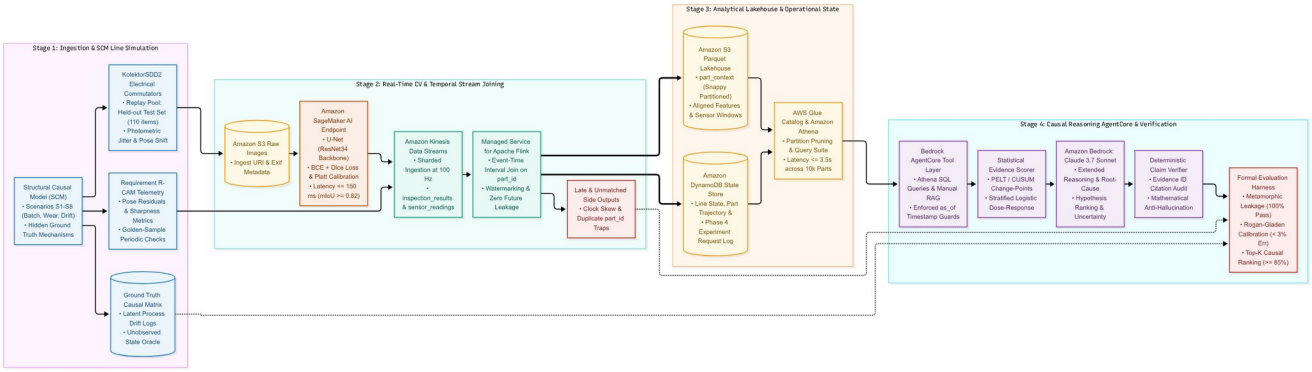


Fig. 2. *End-to-end causal dataflow: SCM simulation, SageMaker AI U-Net segmentation, Managed Service for Apache Flink temporal interval joins, Athena analytical lakehouse, and Bedrock Claude 3.7 Sonnet agent reasoning.*

A. High-Speed Simulation & Computer Vision Pipeline

A Python SCM production simulator emits realistic discrete manufacturing events for electrical commutators, drawing surface defect images from the held-out test split of the KolektorSDD2 dataset (110 positive defect items). High-frequency machine telemetry (100 Hz vibration, spindle current, and feed temperature) is emitted to Amazon Kinesis. Part images uploaded to Amazon S3 trigger an Amazon SageMaker AI real-time endpoint hosting a fine-tuned U-Net with a pre-trained ResNet34 encoder. To prevent overconfident false discoveries, pixel logits are calibrated using Platt scaling, generating predicted defect masks, max-logit flaw scores, and calibrated defect probabilities within 150 ms per part.

B. Temporal Stream Joining & Analytical Lakehouse

Inspection results and sensor telemetry streams converge in Managed Service for Apache Flink. Flink performs an event-time interval join keyed strictly on part_id, bounded by a station transit watermark (200 ms). Unmatched or late-arriving telemetry is routed to dedicated side-output dead-letter queues rather than discarded or incorrectly linked. Aligned part records are written to an Amazon S3 Parquet lakehouse partitioned by date and batch. AWS Glue catalogs the lakehouse schema, enabling Amazon Athena to execute serverless, partition-pruned SQL queries across historical production batches in under 3.5 seconds for 10,000 parts.

C. Bedrock AgentCore & Causal Reasoning Engine

Root-cause investigation is driven by Amazon Bedrock AgentCore executing Anthropic Claude 3.7 Sonnet with extended reasoning capabilities. The agent interacts with the lakehouse exclusively through guarded tools that enforce strict as_of timestamp filtering on both event and ingestion times, guaranteeing zero future-data leakage into retrospective diagnoses. The agent calls statistical scorers (PELT change-point detection, CUSUM drift monitors, and stratified logistic dose-response models) and retrieves standard operating procedures from equipment maintenance manuals via vector search. Diagnostic claims are validated by a deterministic Claim Verifier that requires valid Athena evidence IDs and explicit uncertainty bounds.

III. AWS CLOUD-NATIVE INFRASTRUCTURE & STORAGE ARCHITECTURE

The architecture is engineered exclusively on AWS managed services to ensure horizontal scalability, zero infrastructure management overhead, and strict least-privilege security boundaries. Figure 3 delineates the five architectural tiers:

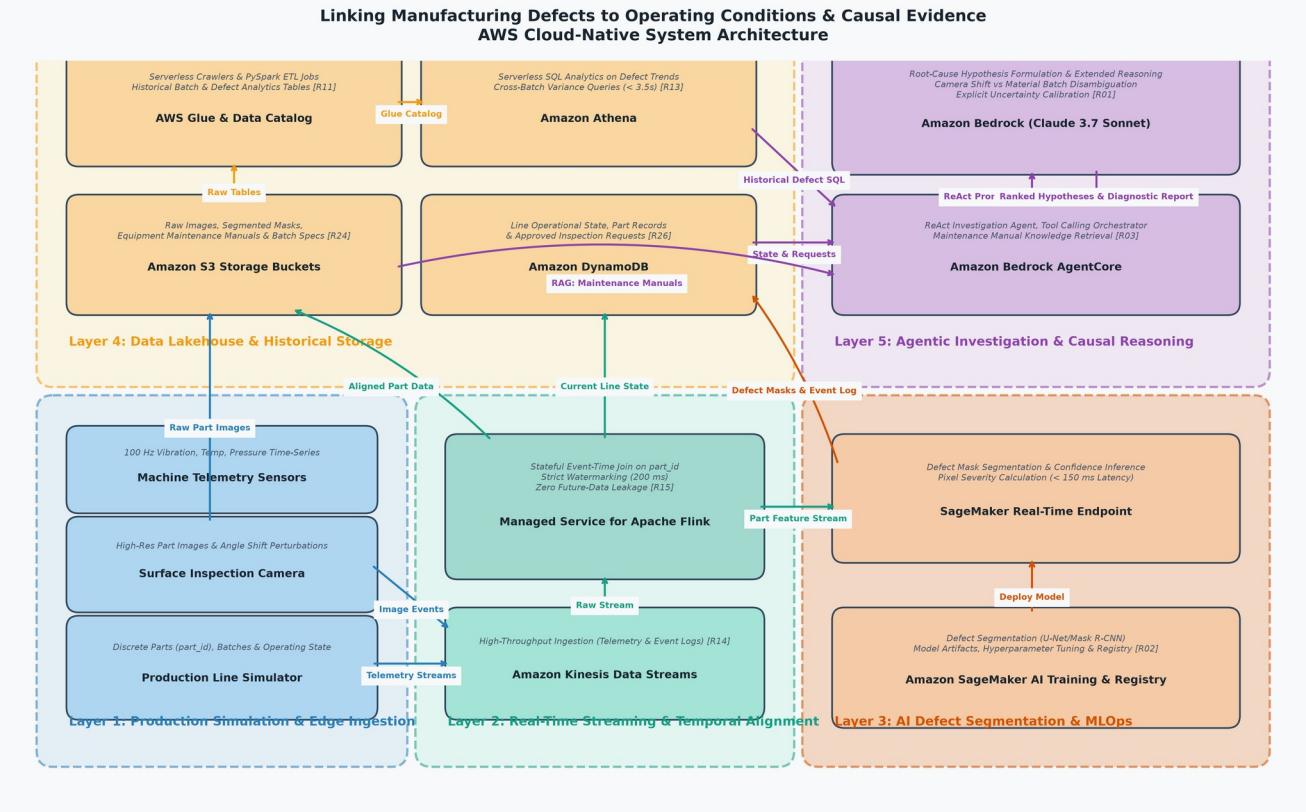


Fig. 3. *Five-layer AWS cloud-native architecture spanning edge simulation, Kinesis/Flink real-time streaming, SageMaker AI MLOps, S3/Athena lakehouse storage, and Bedrock AgentCore reasoning.*

1) Ingestion Layer: High-throughput Amazon Kinesis Data Streams ingest raw 100 Hz machine telemetry and part-inspection events with partition keys assigned to discrete line stations.

2) Stream Processing Layer: Managed Service for Apache Flink maintains managed RocksDB state for watermark tracking, station transit windowing, and temporal interval joining.

3) Model Serving & MLOps: Amazon SageMaker AI hosts containerized model endpoints for defect segmentation, executing batch inference and logging real-time metrics to CloudWatch.

4) Lakehouse & Operational State: Amazon S3 stores raw images, masks, and Snappy Parquet files; Amazon DynamoDB maintains real-time line state, active batch configurations, and the Phase 4 Experiment Request Log.

5) Generative AI & Reasoning: Amazon Bedrock orchestrates Anthropic Claude 3.7 Sonnet within AgentCore, executing automated root-cause investigations with deterministic evidence verification and explicit uncertainty calibration.

IV. FOUR-WEEK SPRINT MILESTONE ROADMAP

The project is structured into four focused, iterative sprints designed to demonstrate rapid incremental capability, early risk retirement, and continuous empirical validation:

TABLE II: 4-WEEK SPRINT MILESTONES & CORE DELIVERABLES

Sprint Week	Sprint Focus	Key Engineering Activities	Verifiable Deliverables
Week 1	Requirements, Simulator & Oracle Baseline	Curate KolektorSDD2 dataset (3,335 items); build SCM line simulator (S1 –S8) with R-CAM optical health channel; establish AWS IAM least-privilege roles; run oracle baseline (E0).	D1: Terraform Cloud IaC, SCM Simulator, S3 Lakehouse & DynamoDB State
Week 2	Segmentation Modeling & Streaming Join	Train U-Net (ResNet34) with BCE+Dice loss (E1) and Platt probability scaling; pipe 100 Hz Kinesis telemetry; deploy Flink event-time interval join (E5).	D2: SageMaker AI Endpoint, Flink Stream Join Job & Partitioned Parquet Lake
Week 3	Lakehouse Cataloging & Agentic Scorer	Implement statistical evidence scorer (PELT/CUSUM); execute R-CAM perturbation test (E7); deploy Bedrock AgentCore tools with as_of guards; build Athena SQL query suite.	D3: Bedrock Claude 3.7 AgentCore, Athena Query Suite & Grounded Tool Layer
Week 4	Perturbation Testing & Final Delivery	Execute metamorphic future-leakage test (E8); perform diagnostic report audit (E9); run dual-perturbation live demo (S1 vs S2); build Phase 4 stretch Experiment Log.	D4: Dual-Perturbation Live Demo, Stretch Experiment Log & Final Report

V. TEAM ALLOCATION, GOVERNANCE & PERFORMANCE TARGETS

The project governance structure assigns specialized ownership across all four team members, with Member 1 (Lead) driving core architectural integrations and the remaining members owning computer vision, lakehouse engineering, and simulation/QA:

TABLE III: TEAM RESPONSIBILITIES & QUANTIFIED PERFORMANCE TARGETS

Member & Role	Specialization	Core Technical Responsibilities	Quantified KPI Targets
Member 1 (Lead)	Principal Architect & Causal Lead	Flink interval joins, Bedrock AgentCore ReAct reasoning, causal ranking logic, SageMaker AI MLOps integration.	Causal Ranking Accuracy >= 85%; Rogan-Gladen rate error < 3%; Zero future-data leakage.
Member 2	Computer Vision Specialist	KolektorSDD2 curation, U-Net ResNet34 training, Platt calibration, SageMaker real-time endpoint deployment.	Defect mIoU >= 0.82; F1-score >= 0.85; Inference Latency <= 150 ms.
Member 3	Data Lakehouse Engineer	S3 Parquet partitioning, AWS Glue ETL crawlers, Athena analytical SQL queries, DynamoDB line state & request store.	Athena Query Latency <= 3.5 s (10k parts); 0% Parquet schema drift.
Member 4	Simulation & QA Lead	SCM line simulator, Kinesis 100 Hz telemetry producer, R-CAM channel, S1–S8 scenario testing, KPI evaluation harness.	Metamorphic Leakage Test Pass Rate = 100%; Telemetry Stream >= 500 events/s.

A. Scope Boundaries & Ethical Safety Guardrails

To preserve rigorous project feasibility within the academic timeline, strict scope boundaries are established: physical factory floor machinery, PLC hardware controllers, and closed-loop actuation are strictly out-of-scope. All production machinery and high-frequency sensors are simulated via our SCM engine. As a Phase 4 stretch extension, the system provides an automated Experiment Request Log in DynamoDB where the agent logs recommended physical experiments to resolve confounded scenarios (e.g., S5), maintaining a transparent human-in-the-loop audit trail.